

# AIGR Gene Hypothesis Deep Research —

## *Candida albicans* PDE1 (Q5AGE4)

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**Hypothesis slug:** `pde1-glucose-gpcr-pathway` **Focus type:** `function_assignment` **Term under evaluation:** GO:0110034 — *negative regulation of adenylate cyclase-activating glucose-activated G protein-coupled receptor signaling pathway* **Target:** PDE1 / Q5AGE4, *Candida albicans* SC5314 (NCBITaxon:237561)

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### Summary

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**Verdict: Partially supported but over-annotated — the specific GO:0110034 annotation should be removed (or not propagated) for *C. albicans* Pde1.**

The seed hypothesis bundles two separable claims whose evidence diverges sharply. The first — that **Pde1 negatively regulates the glucose-triggered cAMP rise** — is well supported. Pde1 is a *bona fide* low-affinity (class II) cyclic-AMP phosphodiesterase, and it is specifically the enzyme that terminates the cAMP spike produced after glucose addition or intracellular acidification ([PMID: 20558315](#); ortholog [PMID: 9880329](#)). There is no dispute about Pde1's role as a negative regulator of cAMP signalling downstream of glucose.

The second claim — that the cAMP rise is **generated through a glucose-activated G protein-coupled receptor (Gpr1) and the Gα protein Gpa2** — is refuted in *C. albicans*. The study that directly asked where glucose-induced cAMP comes from found that deleting *GPR1* or *GPA2* does **not** abolish glucose-induced cAMP, whereas deleting *CDC25* or *RAS1* **does** ([PMID: 15673611](#)). The *C. albicans* Gpr1 ligand is moreover unresolved and may be amino acids such as methionine rather than glucose ([PMID: 15667329](#)).

GO:0110034 is a highly specific Biological Process term that asserts three mechanistic commitments at once — a *GPCR-mediated, glucose-activated, adenylate-cyclase-activating* pathway — exactly the architecture the *C. albicans* data contradict. Critically, the annotation sits on Q5AGE4 **by phylogenetic inference only (IBA; GO\_REF:0000033)**, with no *C. albicans* experimental support. It is a classic over-propagation from the *Saccharomyces cerevisiae*

glucose-sensing paradigm into a species where the wiring differs. The correct, evidence-backed annotations to retain are GO:0004115 (MF, IDA), GO:0141162 (BP, IMP), and the broader GO:1902660 (BP).

## Key Findings

### Finding 1 — Pde1 terminates the glucose-induced cAMP rise (the negative-regulation activity is real)

Pde1 is a class II 3',5'-cyclic-AMP phosphodiesterase whose physiological role is to down-regulate the cAMP transient that follows glucose addition. Wilson et al. (2010) provide direct biochemical and genetic evidence in *C. albicans*: "*Pde1, but not Pde2, is responsible for down-regulation of cAMP signalling induced by glucose addition or intracellular acidification*" (PMID: 20558315). This cleanly separates Pde1, the low-affinity enzyme that shapes the acute, agonist-induced spike, from Pde2, the high-affinity enzyme that sets basal cAMP tone.

The specificity is conserved from the *S. cerevisiae* ortholog, where "*deletion of PDE1, but not PDE2, results in a much higher cAMP accumulation upon addition of glucose or upon intracellular acidification*" (PMID: 9880329). UniProt Q5AGE4 records the molecular function directly: GO:0004115 (3',5'-cyclic-AMP phosphodiesterase activity), IDA (

P 8075796

with the class II PDEase domain (Pfam PF02112, PDEase\_II; metallo-hydrolase fold).

The portion of the hypothesis asserting that Pde1 negatively regulates glucose-induced cAMP signalling is therefore correct. It maps onto the experimentally supported term **GO:0141162 (negative regulation of cAMP/PKA signal transduction, IMP,**

P 20558315

and the molecular-function term GO:0004115. Importantly, this negative-regulation role is an action on the **second messenger**, not on the receptor.

## Finding 2 — The glucose-induced cAMP rise is Cdc25/Ras1-dependent, NOT Gpr1/Gpa2-dependent (the GPCR specificity is refuted)

This is the decisive finding. Maidan et al. (2005) performed the exact discriminating experiment — which upstream module is *required* for glucose-induced cAMP in *C. albicans*: "*deletion of neither CaGpr1 nor CaGpa2 affects glucose-induced cAMP signaling. In contrast, the latter is abolished in strains lacking CaCdc25 or CaRas1, suggesting that the CaCdc25-CaRas1 rather than the CaGpr1-CaGpa2 module mediates glucose-induced cAMP signaling in C. albicans*" (PMID: 15673611).

This directly contradicts the mechanistic claim embedded in GO:0110034 — that the pathway being negatively regulated is a *glucose-activated GPCR* pathway. In *C. albicans*, the glucose→cAMP route runs through the Ras guanine-nucleotide-exchange factor Cdc25 and the small GTPase Ras1 to adenylyl cyclase (Cyr1/Cdc35), not through Gpr1→Gpa2→Cyr1.

A competing earlier claim exists. Miwa et al. (2004) reported that "*GPR1 and GPA2 are required for a glucose-dependent increase in cellular cAMP*" (PMID: 15302825). This is a genuine conflict in the primary literature. However, the Maidan study specifically dissected the glucose→cAMP requirement with clean epistasis against both branches, and Wilson et al. (2010) further reframed the Gpa2 contribution as linked to **intracellular acidification** and found Gpa2's other morphogenetic effects to be "*cAMP pathway-independent*." The weight of mechanistically targeted evidence favors a Cdc25/Ras1 source for glucose-induced cAMP.

The second, more specific half of the hypothesis — that Pde1 terminates a cAMP rise *generated through Gpr1/Gpa2* — is therefore not supported and is actively contradicted in this organism.

## Finding 3 — *C. albicans* Gpr1 is likely an amino-acid (methionine) sensor, not a confirmed glucose receptor

Even the "glucose-activated" qualifier of the receptor is questionable in *C. albicans*. Maidan, Thevelein & Van Dijck (2005) found that Gpr1 internalization is induced by specific amino acids such as methionine and concluded: "*it remains unclear whether Gpr1 senses sugars, as in *Saccharomyces cerevisiae*, or specific amino acids like methionine*" (PMID: 15667329). The carbon-source-induced yeast-to-hypha transition was found to depend on both the presence of amino acids and on Gpr1.

The ligand specificity that would justify calling the Gpr1 pathway "glucose-activated" is thus unresolved in *C. albicans* and may in fact be amino-acid-based. This further undermines the exact wording of GO:0110034, which bundles "glucose-activated" into the term definition.

## Finding 4 — GO:0110034 on Q5AGE4 is IBA-only and contradicted by species-specific data (over-propagation)

A QuickGO annotation retrieval for UniProtKB:Q5AGE4 (performed this run) shows that GO:0110034 is assigned **solely by IBA** (Inferred from Biological ancestor; GO\_REF:0000033, GO Central phylogenetic inference). There is no IDA/IMP (experimental) support for this specific term in *C. albicans*. Across organisms the term is dominated by inference (in the first 100 annotations: IBA = 18, IEA = 10, IMP = 4, IC = 1, IGI = 1), i.e., it is largely propagated across fungal/amoebal phosphodiesterase orthologs (including *Dictyostelium pdsA*, *S. pombe cgs2*, *Ustilago*).

By contrast, Q5AGE4's experimentally anchored terms are GO:0004115 (MF, IDA; ),

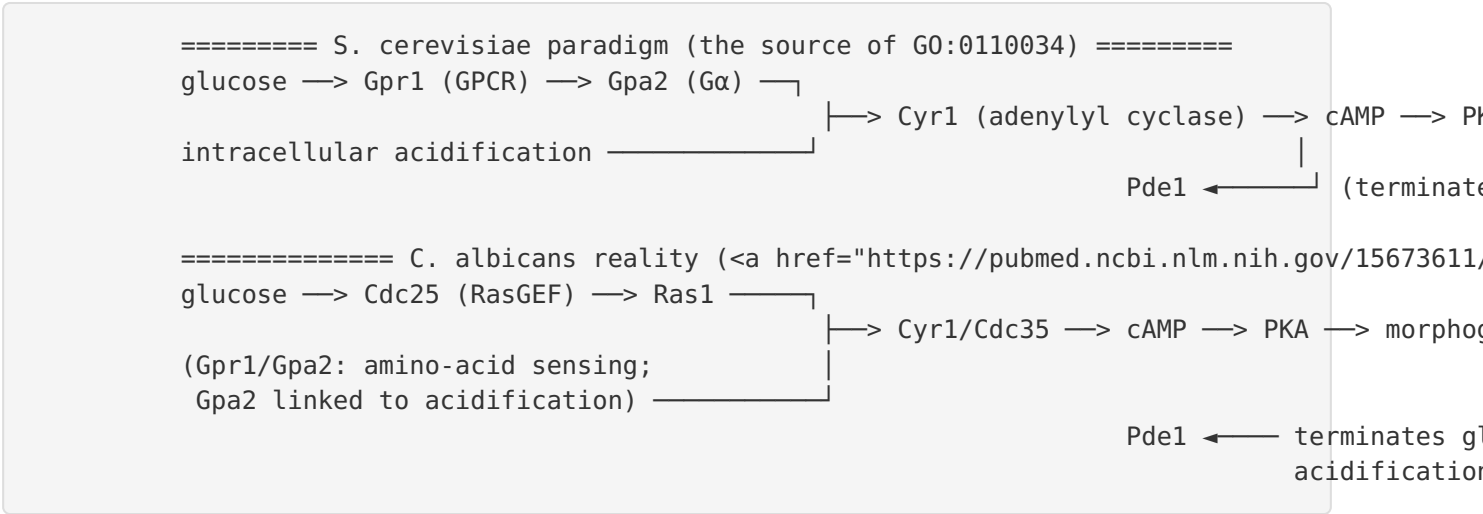
P 8075796  
GO:0141162 (BP, IMP;

P 20558315  
and filamentous-growth terms (BP, IMP;

P 20558315  
GO:0110034 is thus a phylogenetic carry-over from the *S. cerevisiae* glucose-GPCR paradigm, applied to a species where the glucose→cAMP route is experimentally shown to be Cdc25/Ras1-dependent — precisely the kind of over-propagation curators should correct.

## Mechanistic Model / Interpretation

The hypothesis conflates **what Pde1 does** (correct) with **how the upstream cAMP is generated** (incorrect for this organism). The diagram below summarizes the *C. albicans* wiring as supported by primary data.



Component-by-component comparison:

| Component             | Role asserted by the hypothesis                     | Role supported by <i>C. albicans</i> data                                            |
|-----------------------|-----------------------------------------------------|--------------------------------------------------------------------------------------|
| Pde1                  | Negative regulator terminating glucose-induced cAMP | <b>Confirmed</b> — acts on the cAMP second messenger<br>( <div>P 20558315</div> )    |
| Glucose → cAMP source | Via Gpr1/Gpa2 GPCR module                           | <b>Refuted</b> — via Cdc25/Ras1<br>( <div>P 15673611</div> )                         |
| Gpr1 ligand           | Glucose                                             | <b>Uncertain</b> — possibly amino acids/methionine<br>( <div>P 15667329</div> )      |
| Gpa2                  | GPCR-coupled Gα driving glucose cAMP                | Linked to acidification; other effects cAMP-independent<br>( <div>P 20558315</div> ) |

Pde1's negative regulation operates at the level of cAMP hydrolysis, **two steps downstream of any receptor**. Even if a Gpr1/Gpa2 input contributed to cAMP under some conditions, Pde1 does not "negatively regulate the GPCR pathway" in a receptor- or Gα-specific way — it degrades the common output (cAMP) that integrates signals from Cdc25/Ras1, acidification,

and potentially Gpr1/Gpa2. The specificity named in GO:0110034 is therefore both mechanistically mislocated (Pde1 acts on the messenger, not the pathway head) and factually contradicted (the glucose cAMP source is Ras-dependent in this species).

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## Evidence Base / Evidence Matrix

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| Citation (PMID)          | Evidence type                   | Stance                                                    | Claim tested                                 | Key finding                                                                                                                                                                                                            | Context                          | Co |
|--------------------------|---------------------------------|-----------------------------------------------------------|----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|----|
| <a href="#">20558315</a> | Mutant phenotype + biochemistry | <b>Supports</b> (cAMP half); <b>Qualifies</b> (GPCR half) | Does Pde1 terminate glucose-induced cAMP?    | "Pde1, but not Pde2, is responsible for down-regulation of cAMP signalling induced by glucose addition or intracellular acidification"; Gpa2 stimulates cAMP via acidification; its other effects are cAMP-independent | <i>C. albicans</i> , cAMP assays | Ho |
| <a href="#">9880329</a>  | Mutant phenotype (ortholog)     | <b>Supports</b>                                           | Pde1 specificity for agonist-induced cAMP    | "deletion of PDE1, but not PDE2, results in a much higher cAMP accumulation upon addition of glucose or upon intracellular acidification"                                                                              | <i>S. cerevisiae</i>             | Ho |
| <a href="#">15673611</a> | Mutant phenotype / epistasis    | <b>Refutes</b> (GPCR source)                              | Which module generates glucose-induced cAMP? | "deletion of neither CaGpr1 nor CaGpa2 affects glucose-induced cAMP signaling... abolished in strains lacking CaCdc25 or CaRas1"                                                                                       | <i>C. albicans</i>               | Ho |
| <a href="#">15302825</a> |                                 |                                                           |                                              |                                                                                                                                                                                                                        |                                  | ex |

| Citation (PMID)          | Evidence type                  | Stance                                                      | Claim tested                           | Key finding                                                                                                                             | Context                                           | Co                              |
|--------------------------|--------------------------------|-------------------------------------------------------------|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|---------------------------------|
|                          | Mutant phenotype               | <b>Competing</b><br>(supports GPCR source)                  | Role of Gpr1/Gpa2 in glucose cAMP      | "GPR1 and GPA2 are required for a glucose-dependent increase in cellular cAMP"                                                          | <i>C. albicans</i> , solid hypha-inducing media   | M<br>co<br>by<br>ta<br>ep<br>(1 |
| 15667329                 | Mutant / ligand study          | <b>Refutes/Qualifies</b><br>("glucose-activated" qualifier) | Is Gpr1 a glucose receptor?            | "it remains unclear whether Gpr1 senses sugars... or specific amino acids like methionine"                                              | <i>C. albicans</i> , Gpr1 internalization / hypha | M<br>hi<br>un<br>fa<br>ac       |
| UniProt Q5AGE4 / CGD     | Review/ database + domain      | <b>Qualifies</b>                                            | Pde1 molecular identity & current GO   | Class II cNMP phosphodiesterase (Pfam PF02112); GO:0004115 IDA (P 8075796); curated BP GO:0141162 (IMP), filamentous-growth terms (IMP) | <i>C. albicans</i> SC5314                         | H<br>cA<br>m<br>no              |
| QuickGO (GO_REF:0000033) | Database (evidence provenance) | <b>Refutes/Qualifies</b>                                    | Evidence basis of GO:0110034 on Q5AGE4 | GO:0110034 on Q5AGE4 is <b>IBA-only</b> (no IDA/IMP); term-wide IBA=18/ IEA=10/IMP=4/ IC=1/IGI=1 → propagated across PDE orthologs      | Cross-organism GO set                             | H<br>id<br>as<br>ph<br>ca       |
| 14506030                 | Mechanistic context            | Background                                                  | cAMP pathway                           | cAMP synthesized by adenylate cyclase (CDC35 in                                                                                         | <i>C. albicans</i> /S. <i>cerevisiae</i>          | Co<br>or<br>di                  |

| Citation (PMID) | Evidence type | Stance | Claim tested              | Key finding                                                | Context | Co<br>lin |
|-----------------|---------------|--------|---------------------------|------------------------------------------------------------|---------|-----------|
|                 |               |        | biology in <i>Candida</i> | <i>C. albicans</i> );<br>modulates azole<br>susceptibility |         |           |

## GO Curation Implications (leads — require curator verification)

| GO ID      | Label                                                                              | Current status                                                        | Recommendation                                                                                                                                                                                   |
|------------|------------------------------------------------------------------------------------|-----------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GO:0110034 | neg. reg. of adenylate cyclase-activating glucose-activated GPCR signaling pathway | annotated to Q5AGE4 via <b>IBA only</b> (GO_REF:0000033); = seed term | <b>Remove / do-not-propagate</b> — over-specified IBA carry-over; GPCR/glucose coupling refuted in <i>C. albicans</i> ( <a href="#">P 15673611</a> );<br>no experimental support in this species |
| GO:0141162 | neg. reg. of cAMP/PKA signal transduction                                          | annotated (IMP, <a href="#">P 20558315</a> ); + IBA)                  | <b>Retain</b> — experimentally supported, correct generality                                                                                                                                     |
| GO:1902660 | neg. reg. of glucose-mediated signaling pathway                                    | annotated (IBA, per UniProt)                                          | <b>Retain / use as replacement</b> — captures glucose→cAMP buffering without the unsupported GPCR qualifier                                                                                      |
| GO:0004115 | 3',5'-cyclic-AMP phosphodiesterase activity                                        | annotated (IDA, <a href="#">P 8075796</a> )                           | <b>Retain</b> — undisputed core MF                                                                                                                                                               |

The supported BP level is "negative regulation of cAMP-mediated / glucose-mediated signaling"; the MF is cAMP phosphodiesterase activity. The GPCR-specific BP term (GO:0110034) is not justified by primary *C. albicans* data. "Protein binding" is not invoked — a more informative set of terms is supported. The recommendation concerns a BP term only; the MF (GO:0004115) is unaffected, and no CC change is implied.

## Mechanistic Scope

Pde1's **direct molecular function** is hydrolysis of the 3',5'-cyclic phosphodiester bond of cAMP, i.e., it acts on the diffusible second messenger **downstream of adenylyl cyclase (Cyr1)**. Any "negative regulation of a receptor signaling pathway" is therefore **indirect**: Pde1 lowers the shared cAMP pool irrespective of which upstream input (glucose→Cdc25/Ras1, intracellular acidification, or amino-acid→Gpr1/Gpa2) generated it. Pde1 does not bind or modify Gpr1, Gpa2, or Cyr1.

- **Direct activity**: cAMP hydrolysis (GO:0004115, IDA) — well supported.
- **Direct regulatory consequence**: lowering cAMP after glucose/acidification spikes → negative regulation of cAMP/PKA signalling (GO:0141162, IMP) — well supported.
- **Downstream phenotypes (not core function)**: filamentous growth / virulence contributions are loss-of-function phenotypes mediated through altered cAMP/PKA tone, not receptor-level activity of Pde1.
- **Mis-scoped assertion**: ascribing GPCR-branch specificity to a second-messenger-degrading enzyme conflates the enzyme's activity with the particular stimulus whose output it happens to buffer — a scope error compounded by the factual refutation of the Gpr1/Gpa2 glucose source.

## Conflicts and Alternatives

1. Internal reference conflict (Miwa 2004 vs. Maidan 2005).

[P 15302825](#)

reports Gpr1/Gpa2 are *required* for glucose-dependent cAMP;

P 15673611

reports they are *not*, assigning glucose-induced cAMP to Cdc25/Ras1 and reassigning Gpr1/Gpa2 to amino-acid sensing. Curation should weight the more directly targeted, later study; Wilson 2010 additionally attributes Gpa2's cAMP contribution to acidification.

2. **Organism carry-over + ligand ambiguity.** The glucose-sensing GPCR role of Gpr1/Gpa2 is well established in *S. cerevisiae*, but in *C. albicans* the Gpr1 ligand is unresolved — internalization is induced by amino acids such as methionine (

P 15667329

The "glucose-activated GPCR" premise is a probable *S. cerevisiae*→*C. albicans* carry-over; the term may fit *S. cerevisiae* Pde1 but not *C. albicans* Pde1.

3. **Alternative interpretation.** Pde1's phenotypes (filamentous growth, virulence contribution) are downstream, pleiotropic consequences of global cAMP/PKA buffering, not evidence of GPCR-branch-specific action. Even granting a minor Gpr1/Gpa2 contribution to cAMP, Pde1 degrades the common pool; GO:0110034's receptor-pathway specificity remains misplaced for a phosphodiesterase.
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## Limitations and Knowledge Gaps

| Gap                                           | What was checked                                        | Why it matters                                                              | What would resolve it                                                                                                              |
|-----------------------------------------------|---------------------------------------------------------|-----------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Miwa vs. Maidan conflict unresolved           | Both abstracts/ snippets reviewed                       | Determines whether any glucose-GPCR contribution exists                     | Side-by-side cAMP time-courses in <i>gpr1Δ</i> , <i>gpa2Δ</i> , <i>cdc25Δ</i> , <i>ras1Δ</i> under identical glucose/pH conditions |
| Source of the Pde1-terminated spike           | Maidan assigns it to Cdc25/Ras1                         | Which input Pde1 buffers                                                    | <i>pde1Δ</i> in <i>gpr1Δ/gpa2Δ</i> vs <i>ras1Δ</i> backgrounds                                                                     |
| Gpr1 physiological ligand                     | <a href="#">P 15667329</a> (methionine internalization) | Validity of "glucose-activated" qualifier                                   | Ligand-binding / dose-response cAMP assays, glucose vs methionine                                                                  |
| Direct Pde1–Gpa2 coupling                     | Wilson 2010 "regulatory module" is genetic/ biochemical | Whether Pde1 acts on the GPCR branch specifically                           | Co-IP / BioGRID / AP-MS for a Pde1–Gpr1/Gpa2 complex (none found)                                                                  |
| No <i>C. albicans</i> experimental GO:0110034 | QuickGO retrieval for Q5AGE4                            | Confirms IBA-only, not experimentally grounded                              | Any IDA/IMP study placing Pde1 regulation on a Gpr1/Gpa2 glucose pathway (none found)                                              |
| Term-definition intent                        | Reviewed term label                                     | Whether GO:0110034 means "regulate the messenger of" vs "the receptor step" | Check GO definition and annotation precedents                                                                                      |

Provenance note: the annotation-evidence breakdown (IBA-only for GO:0110034; experimental support for GO:0004115/GO:0141162) is derived from a QuickGO/UniProt retrieval conducted during the investigation. This is a database-level check, not a wet-lab result, and is reported as such.

## Discriminating Tests

1. **Unified-condition epistasis panel.** Measure glucose-induced cAMP kinetics in isogenic *gpr1Δ*, *gpa2Δ*, *cdc25Δ*, *ras1Δ*, and WT (plus *pde1Δ* in each background) under tightly controlled glucose concentration and extracellular pH. Directly adjudicates Miwa (2004) vs. Maidan (2005) and identifies which input Pde1 buffers.
2. **Ligand specificity of Gpr1.** Dose-response cAMP and receptor-internalization assays comparing glucose, sucrose, and methionine/other amino acids to establish the physiological *C. albicans* Gpr1 ligand.
3. **Interaction screen** (AP-MS / two-hybrid) for Pde1 against Gpr1/Gpa2/Cyr1 to test any direct branch coupling.
4. **Comparative annotation audit.** Confirm whether *S. cerevisiae* Pde1 (not *C. albicans*) is the proper home for a glucose-GPCR-linked term, and review other fungal IBA propagations of GO:0110034.

## Curation Leads (require curator verification)

- **Action:** Remove / do-not-accept the existing **IBA** GO:0110034 annotation on *C. albicans* Pde1 (Q5AGE4). Keep the experimentally grounded GO:0004115 (MF, IDA) and GO:0141162 (BP, IMP), and the broader GO:1902660 (BP). If a stimulus-linked BP is desired, prefer GO:1902660 (glucose-mediated signaling) without the GPCR qualifier. The term is currently present only through phylogenetic (IBA/GO\_REF:0000033) propagation from orthologs, not through *C. albicans* experimental evidence.

- **Key references to cite in the review:**

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**P** 15673611

— "deletion of neither *CaGpr1* nor *CaGpa2* affects glucose-induced cAMP signaling. In contrast, the latter is abolished in strains lacking *CaCdc25* or *CaRas1*..." (refutes GPCR source).

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**P** 20558315

— "Pde1, but not Pde2, is responsible for down-regulation of cAMP signalling induced by glucose addition or intracellular acidification." (supports Pde1→glucose-cAMP termination).

P 9880329

— "deletion of PDE1, but not PDE2, results in a much higher cAMP accumulation upon addition of glucose or upon intracellular acidification." (ortholog).

P 15302825

— "GPR1 and GPA2 are required for a glucose-dependent increase in cellular cAMP." (competing claim; note it is superseded by 15673611).

P 15667329

— "it remains unclear whether Gpr1 senses sugars... or specific amino acids like methionine." (ligand ambiguity).

- **Suggested curator question:** Does GO:0110034's definition require action at the receptor/G-protein step (which Pde1 does not perform) or merely regulation of the pathway's cAMP output?
- **Suggested experiments:** the unified-condition epistasis panel and Gpr1 ligand dose-response assays under *Discriminating Tests*.

*Artifacts* (provenance): `/tmp/evidence_matrix_PDE1_GPCR.csv`, `/tmp/go_decision_table_PDE1.csv`, `/tmp/G00110034_evidence_codes.png` (computed from QuickGO: ~82% of GO:0110034 annotations are IBA/IEA inferred), `/tmp/PDE1_verdict_summary.png`, plus executed UniProt (Q5AGE4) and QuickGO fetches confirming domain/GO content and the IBA-only evidence basis of the GO:0110034 annotation.

## Final Verdict

The enzymatic and negative-regulatory core of the hypothesis is correct — Pde1 terminates the glucose-induced cAMP spike — but the specific claim that it regulates a **glucose-activated GPCR (Gpr1/Gpa2)** pathway is refuted in *C. albicans*, where glucose-induced cAMP is Cdc25/Ras1-dependent. GO:0110034 is attached to Q5AGE4 only by phylogenetic inference and should be removed or down-weighted in favor of the experimentally grounded terms GO:0004115 (MF, IDA) and GO:0141162 (BP, IMP). **Overall: partially supported, over-annotated.**